Lesson plan · for the teacher

From sheets to a database

Data engineering · Level 3 · Lesson 1

Time
40 min

Materials
Board and chalk; A wide made-up school table, drawn or printed; Optional: a phone or computer for the SQL steps

Learning goals

- Say what a database adds to a spreadsheet.
- Store each fact once, in tables linked by keys.
- Read a CREATE TABLE and run a first SELECT.

Activity: One fact, one place

1. Warm-up (5 min): Draw the wide table on the board and ask pupils to count how many times each teacher's phone number appears.
2. Teach (8 min): Explain one fact in one place, a primary key and a foreign key, using the pupils and classes as the example.
3. Split (15 min): Groups redraw the wide table as a pupils table and a classes table, and mark the primary key and the foreign key in each.
4. Change (7 min): Change one teacher's phone in both designs and count the cells each one needed. Then remove the last pupils of one class: what disappears?
5. Wrap-up (5 min): Show the CREATE TABLE for the classes table and a first SELECT. Then give the exit ticket.

Use made-up names and phone numbers in the wide table. Make at least one class have several pupils, so the repeats are easy to see.

Discussion

- What went wrong when we changed the phone in the wide table?
- Which column links the pupils table to the classes table?
- Why is a phone number stored as text?

Exit ticket

- Name a primary key and a foreign key in our two tables, and say what each one does.

Try it at school tomorrow
Draw one wide table on the board, with pupils, their class, the class teacher and her phone. Pupils split it into two tables with a key. Then change the phone number: how many places did each design need?

Practice questions · for you

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1 **Each mark row stores the pupil's number, pointing to a row in the pupils table. What is that column?**

◯ A foreign key ◯ A formula ◯ A primary key

2 **Match each word to what it means in a school database.**

Table	• • • •	The pupil ID that picks out one pupil's row
Primary key	• • • •	Rows of one kind of thing, such as pupils or classes
Foreign key	• • • •	Whole number for a mark, text for a name
Type	• • • •	The pupil ID inside a mark row

3 **Put the lines in order to create the classes table, with its columns in this order: class (the key), teacher, phone.**

Number the lines in order (1, 2, 3…), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐ `class TEXT PRIMARY KEY,`

☐ `CREATE TABLE classes (`

☐ `phone INTEGER`

☐ `);`

☐ `phone TEXT`

☐ `teacher TEXT,`

4 **To make reports quicker, each pupil's name should also be stored in this marks table.**

```
CREATE TABLE marks (
    pupil_id INTEGER REFERENCES pupils(id),
    subject TEXT,
    mark INTEGER
);
```

◯ True ◯ False

5 **Your first query. Change it so it returns only two columns for every pupil: their name and their class. Press Run to see the tables first.**

```
SELECT * FROM pupils;
```

Write your program:

✎ **Which fact does your school type in more than one place? Which table should it live in, once?**

Check your answers

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1 **A foreign key**

A foreign key is a column that points to a row in another table, where that value is the primary key.

2 **Table — Rows of one kind of thing, such as pupils or classes**
Primary key — The pupil ID that picks out one pupil's row
Foreign key — The pupil ID inside a mark row
Type — Whole number for a mark, text for a name

Tables hold one kind of thing each. The primary key identifies a row in its own table; a foreign key points to that row from another table; and types stop a name landing in a mark column.

3 **Top to bottom: 2, 1, X, 5, 4, 3**

```
CREATE TABLE classes (
    class TEXT PRIMARY KEY,
    teacher TEXT,
    phone TEXT
);
```

CREATE TABLE names the table, then each column with its type, separated by commas, with no comma after the last.

4 **False**

Store each fact once. The name already lives in the pupils table; pupil_id points to it, and a query can bring the name in when it's needed.

5

```
SELECT name, class FROM pupils;
```

SELECT * means every column; listing name, class asks for just those two.